Engineering Log - Day 13

Adaptive SDR Spectrum Analyzer / Cognitive RF Decision System

Project	Adaptive SDR Spectrum Analyzer
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Session Date	July 20, 2026
Session Focus	Reliability, measurement quality, adaptive detection, decision confidence, verification, and report UI
Development Method	One controlled change at a time: implement, test, inspect live RTL-SDR output, commit, and push
Status at End of Session	Stable Day 13 baseline; 65 automated tests passing; professional survey report design established

1. Executive Summary

Day 13 transformed the analyzer from a working prototype into a substantially more defensible engineering system. The most important architectural change moved RTL-SDR acquisition and frequency tuning away from the Qt GUI thread. This eliminated the original application-freeze failure mode and introduced a controlled worker interface for sample delivery, tune requests, error reporting, and shutdown.

The survey path was then stabilized with post-tune frame buffering, validation, and median aggregation. Detection was upgraded from a fixed global threshold to a resolution-aware, locally adaptive pipeline using a gain-compensated Hann FFT and temporal peak confirmation. The SMART recommendation engine was normalized, made deterministic, and extended with runner-up, decision-margin, and score-separation confidence reporting.

Day 13 also established an automated regression baseline of 65 passing tests and introduced the first production-quality UI component: a structured Survey Analysis Report. This report is the visual reference for the rest of the application.

2. Starting Point

Area	Starting Condition	Engineering Risk
SDR I/O	Read and tune operations executed in the GUI path	USB delays or failures could freeze the application
Survey samples	A single fresh measurement could decide a scan point	Noise spikes and tune transients could bias results
Peak detection	Global mean threshold and fixed-bin separation	Behavior changed with FFT resolution and curved baselines
SMART scoring	Useful but partly absolute and insertion-order dependent	Scores were difficult to compare and ties were ambiguous
Verification	Primarily manual live-hardware testing	Regression risk increased with every subsystem change
Results popup	Large monospaced text block	Correct data but limited hierarchy and presentation quality

3. Day 13 Engineering Objectives

- Protect the Qt event loop from blocking RTL-SDR operations.
- Make survey measurements repeatable across tune transitions.
- Improve peak detection without hiding narrow or weak signals.
- Make SMART results deterministic, bounded, and explainable.
- Add regression tests before expanding functionality further.
- Establish a professional UI language without destabilizing live rendering.

4. Non-Blocking RTL-SDR Architecture

The central reliability milestone was the introduction of SDRWorker, a QThread that owns SDRManager. The worker performs continuous sample reads and processes tune requests outside the main Qt event loop. Signals return samples and status changes to the GUI thread, where plotting and widget updates remain safe.

```
GUI thread                           SDR worker thread
-----------                          -----------------
Tune button  -- request_tune() ----> pending frequency
                                      |
                                      +--> SDRManager.tune()
                                      +--> SDRManager.read_samples()
                                      |
FFT / UI      <-- samples_ready -----+
Status panel  <-- tune/error signals-+
```

Thread-safe tuning

A lock protects the pending-frequency request. The worker consumes the newest request, performs the hardware operation, and emits either tune_succeeded or tune_failed. The displayed center frequency is updated only after hardware success.

```
def request_tune(self, freq_hz):
    if not self.isRunning():
        return False
    with self._tune_lock:
        self._pending_frequency = float(freq_hz)
    return True
```

Input and hardware-failure handling

Condition	System Response	Result
Blank, text, NaN, or infinite frequency	Reject input and preserve the last valid center frequency	No invalid tuner command
Center outside RTL-SDR V3 operating range	Display a bounded validation error	Hardware remains unchanged
Tune operation fails	Emit failure and retain the last confirmed frequency axis	No false-success display
USB sample read returns an error	Clear measurement availability and report SDR failure	Survey terminates safely
Window closes during survey	Stop survey timer, request worker interruption, wait for exit	Clean process shutdown

The RTL-SDR library can still report low-level USB or tuner errors such as LIBUSB_ERROR_OTHER. Day 13 does not attempt to hide these hardware faults. It ensures they are surfaced clearly and cannot leave the GUI frozen, the survey timer active, or the displayed frequency out of sync with the dongle.

5. Survey Lifecycle and Shutdown Safety

Survey state is now reset explicitly at the start of a run. A second Start Survey command is rejected while a survey is active, and Clear Survey removes results, metrics, heatmap state, recommendation output, and cached popup content as one operation.

Lifecycle Event	Required State Transition
Start	Reset stale state, validate inputs, enter active scan
Tune step	Invalidate old measurement, request worker tune
Measurement error	Stop timer and report unavailable RF data
Completion	Rank candidates, render report, retain results
Application close	Stop survey, interrupt worker, wait, exit

6. Post-Tune Measurement Stabilization

A survey point should represent RF conditions after the tuner has settled, not the first frame delivered during a transition. Day 13 introduced a bounded measurement buffer and a validated median aggregator. Each accepted frame contributes occupancy, maximum power, and average power.

```
Tune confirmed
    |
    +--> clear previous measurement buffer
    +--> collect post-tune frames
    +--> validate finite numeric fields
    +--> require at least 3 frames
    +--> median(occupancy, max_power, average_power)
    +--> save one survey point
```

Why the median was selected

The median rejects a single-frame spike without requiring a complex statistical model. It is computationally inexpensive, easy to explain, and well matched to a small post-tune sample window. The current buffer holds up to 10 recent frames and the survey requires at least three valid observations.

Measurement Set	Mean Occupancy	Median Occupancy	Interpretation
8.1, 8.4, 35.0	17.2%	8.4%	Median rejects the isolated tune transient
9.0, 9.2, 9.4	9.2%	9.2%	Both estimators agree for stable frames
Only two valid frames	Not used	Not used	Survey waits because confidence is insufficient

7. Survey Display Stability

The status panel previously flickered between normal live status and transient tuning messages as the survey moved from frequency to frequency. The status-update path is now gated while a tune is pending, and automated survey tunes can suppress the temporary manual-tune message. This preserves a stable display while still reporting real errors.

Display Element	Day 13 Behavior
Center-frequency heading	Changes only after tune success
Status panel	Does not overwrite active tune or error state
FFT and waterfall axes	Move from the confirmed center frequency
Survey progress	Advances only after a valid aggregate is recorded
Heatmap recommendation marker	Drawn from the completed recommendation

8. Frequency-Axis and Image Alignment

Waterfall and heatmap rectangles now use frequency-bin edges rather than bin centers. Extending the image by half a bin on both sides aligns raster pixels with plotted FFT coordinates and prevents visible axis drift.

```
bin_width = freqs_mhz[1] - freqs_mhz[0]
left_edge = freqs_mhz[0] - bin_width / 2
right_edge = freqs_mhz[-1] + bin_width / 2
```

This correction is small in code but important in measurement software: a displayed RF feature and its numeric frequency must refer to the same physical bin.

9. Adaptive Detection Pipeline

Day 13 replaced several resolution-dependent assumptions with physical-frequency parameters and local statistics. The resulting pipeline is more stable across FFT sizes and across the curved front-end response commonly observed with an RTL-SDR V3.

```
Complex IQ samples
      |
Gain-compensated Hann FFT
      |
Local 30th-percentile noise floor (250 kHz window)
      |
Per-bin threshold = local floor + 10 dB
      |
Resolution-independent peak separation (75 kHz)
      |
Bandwidth measurement from FFT bin spacing
      |
Temporal confirmation: 2 hits in 3 frames within 25 kHz
      |
Confirmed peaks -> UI, history, FeatureStore, survey scoring
```

Physical peak separation

The previous fixed distance of 300 FFT bins represented different RF spacing whenever sample count or sample rate changed. Day 13 converts a 75 kHz minimum separation into bins at runtime.

```
bin_width_khz = abs(freqs_mhz[1] - freqs_mhz[0]) * 1000
distance_bins = max(1, round(75.0 / bin_width_khz))
```

Local adaptive noise floor

A local 30th-percentile estimator follows slow baseline curvature while remaining below narrow carriers. The detector adds a 10 dB margin to produce a threshold for every FFT bin. This prevents a high-noise region on one side of the spectrum from suppressing a valid signal elsewhere.

Detector Property	Day 13 Configuration	Engineering Purpose
Noise window	250 kHz	Track slow RF baseline variation
Noise percentile	30th	Reject narrow peaks from floor estimate
Detection margin	10 dB	Require clear local prominence
Peak spacing	75 kHz	Prevent duplicate detections of one feature
Confirmation	2 of 3 frames	Reject single-frame impulses
Match tolerance	25 kHz	Allow small frequency drift

Gain-compensated Hann FFT

A Hann window reduces spectral leakage for off-bin tones. Coherent gain compensation preserves the amplitude of a bin-centered tone, so improved leakage behavior does not silently change the existing power scale.

Measured Effect	Observed Result
Off-bin leakage reduction	Greater than 15 dB in regression testing
Additional processing time	Approximately 0.086 ms per frame
Local-floor processing time	Approximately 0.21 ms per 8192-bin frame
Temporal confirmation overhead	Approximately 6.9 microseconds per frame

10. Rejected Global-Threshold Experiment

A global mean/MAD threshold was evaluated before the local-floor design was adopted. It appeared reasonable on a flat spectrum but failed during live testing near 120 MHz. The RTL-SDR response was curved enough that the global threshold rose to approximately 48.5 dB, producing zero detected signals and zero occupancy even while visible carriers remained on the FFT.

Approach	Live Observation	Decision
Global mean plus fixed margin	Sensitive to broad baseline elevation	Replaced
Global median/MAD candidate	Still represents one threshold for a nonuniform spectrum	Rejected after evaluation
Local percentile floor	Tracks baseline by frequency and retains narrow peaks	Integrated

Engineering lesson: the threshold must model the receiver baseline as a function of frequency. A statistically robust global value is still the wrong model when the baseline itself is not uniform.

11. Temporal Peak Confirmation

Peak confirmation maintains a rolling three-frame detection window. A signal is published after two frequency-consistent hits within 25 kHz. One missed frame is tolerated, but a single isolated spike never reaches the signal table, history, or FeatureStore.

Frame Sequence	Expected Result
hit, hit	Confirmed on the second observation
hit, miss, hit	Confirmed; one missed frame is tolerated
hit, miss, miss	Expired; no published signal
hit at f, hit more than 25 kHz away	Treated as separate candidates
Tune to new center frequency	Confirmation history reset

12. Signal-State Freshness and Memory Hygiene

Signal features and history entries now track last_seen using a monotonic clock. Entries older than 30 seconds are pruned during live processing. FeatureStore keys are normalized to 0.01 MHz so minor peak jitter does not create unbounded duplicate records.

State Concern	Implemented Control	Effect
Peak-frequency jitter	Normalize keys to 0.01 MHz	Stable identity for nearby detections
Tune away and return later	Prune after 30 seconds	Old age does not leak into a new observation

State Concern	Implemented Control	Effect
Mutable global features during survey	Capture a feature snapshot at each survey point	Recommendation is based on the measured scan state
Repeated UI updates in one cycle	Count a signal once per update cycle	Persistence is not inflated

13. SMART Decision Engine Normalization

The SMART engine remains deliberately explainable. Day 13 did not replace it with a black-box model. Instead, each score component was bounded, configured centrally, and made comparable across survey runs.

Component	Maximum	Day 13 Interpretation
Occupancy	50	Maps 0-100% occupancy across the full 50-0 range
Relative power	30	Median-centered and insensitive to absolute offset
Persistence	15	Rewards established N/A/P/L signal behavior
Age	10	Rewards sustained observation without dominating
Strength	6	Small W/M/S classification contribution
Total SMART maximum	111	Single configured denominator for reports

Offset-invariant power scoring

Absolute dB values from an inexpensive SDR are affected by gain, front-end response, and calibration. Power scoring is therefore centered on the survey median. Adding the same offset to every candidate does not change their relative power scores.

Deterministic selection

Candidates are sorted with an explicit tie-break policy rather than relying on dictionary insertion order. Identical inputs now produce the same recommendation regardless of how results were constructed.

Priority	Tie-Break Field	Purpose
1	Highest overall SMART score	Primary optimization target
2	Lower occupancy	Prefer the cleaner equal-score candidate
3	Lower center frequency	Provide stable final ordering

14. Runner-Up, Margin, and Confidence

The recommendation now records the second-ranked candidate and the decision margin. This exposes whether the winner was dominant or nearly tied. Confidence is calculated from normalized score separation, not from an unsupported statistical probability.

Reported Field	Meaning
Runner-Up	Second-highest deterministic candidate
Decision Margin	Winner score minus runner-up score
Normalized Separation	Decision margin divided by SMART maximum
LOW / MODERATE / HIGH	Policy labels for score separation
N/A	Used when no runner-up exists

The UI explicitly states: confidence reflects winner/runner-up score separation, not statistical certainty. This wording is important because the system has not yet accumulated the repeated survey population required for a true confidence interval.

15. Decision Output at End of Day 13

```
Recommendation
  frequency, occupancy, score, maximum_score
  runner_up_frequency, runner_up_score
  decision_margin, decision_confidence
  occupancy_score, power_score
  persistence_score, age_score, strength_score
  max_power, average_power, reason
```

16. Automated Regression Baseline

Day 13 established a repeatable unittest suite covering signal processing, state management, survey logic, and recommendation behavior. The final verification run completed 65 tests in 0.006 seconds with no failures.

Test Area	Examples of Verified Behavior
Decision engine	Score maxima, full-range occupancy, offset-invariant power, runner-up, margin, confidence, fallback, deterministic ties
Peak detection	Top-three selection, physical spacing, bandwidth, invalid input
Local noise floor	Flat and curved floors, narrow peaks, physical window, margin validation
FFT processing	Tone bin, amplitude preservation, leakage reduction, finite zero input
Measurement aggregation	Minimum frames, medians, missing fields, invalid values, outlier rejection
Peak confirmation	Two-of-three hits, drift tolerance, expiry, reset, missed-frame tolerance
Feature and history state	Normalized keys, last-seen pruning, update-cycle counting
Frequency and survey utilities	Half-bin edges, occupancy arrays, float-safe scan generation, report labels

Final automated result

```
----------------------------------------------------------------------
Ran 65 tests in 0.006s

OK
```

17. Live RTL-SDR V3 Validation

Automated tests were supplemented by repeated live-hardware checks because USB behavior, tuner settling, and RF baseline shape cannot be represented completely by unit tests.

Manual Validation	Observed Result
Launch and continuous display at 100 MHz	Stable FFT, waterfall, table, and age updates
Manual tune to 120 MHz	Frequency axis and status changed after tune success
Invalid manual input	Blank, text, NaN, and out-of-range values rejected
88-92 MHz survey, 1 MHz step	Five points completed and recommendation rendered
88-92 MHz survey, 0.2 MHz step	Twenty-one points completed without status flicker
Larger 13-point survey	Completed with ranked report output

Manual Validation	Observed Result
Repeated surveys	Three consecutive full runs completed normally
Close during active survey	Worker closed cleanly without Qt warnings
Tune away and return after stale interval	Signal age restarted from zero

This combined test strategy is intentional: unit tests protect deterministic logic, while live RTL-SDR checks validate the actual device, driver, thread, and GUI integration.

18. Professional Survey Report UI

The Survey Results popup became the first component updated to the project's professional UI standard. The previous terminal-like text view was replaced with a structured HTML report inside a scrollable Qt text document.

Visual Layer	Implemented Treatment
Application background	Dark charcoal
Report canvas	Near-black content surface with restrained border
Primary accent	Cyan-blue for the recommendation hero
Semantic status	Green for high confidence; amber/red reserved for caution/error
Typography	White headings, muted gray labels, strong numeric hierarchy
Data presentation	Cards and aligned tables rather than decorative separators
Actions	Consistent blue primary Close button

Report hierarchy

```
Survey Analysis Report
  Summary cards
    - Points scanned
    - Average occupancy
  Recommendation hero
    - Mode and recommended frequency
    - Occupancy and overall score
  Decision comparison
    - Recommended candidate vs runner-up
    - Margin and score-separation confidence
  Score breakdown
  Decision rationale
  Ranked measured occupancy
```

The layout was verified with a live five-point SMART survey. The window supports scrolling for longer reports and retains useful minimum dimensions for smaller displays. The user visually approved the final design and selected it as the reference style for the entire application.

19. Application-Wide Design Standard

Rule	Application-Wide Direction
Color	Use the popup's charcoal, card, cyan-blue, and semantic palette
Spacing	Adopt consistent outer margins, section gaps, and card padding
Hierarchy	Separate title, label, value, status, and supporting text roles
Tables	Use aligned columns, distinct headers, and alternating dark rows
Controls	Use one primary action style and quieter secondary actions
Performance	Avoid live auto-ranging or visual effects that reduce FFT frame rate
Migration	Update one UI region at a time with a live-hardware check after each change

UI modernization is now authorized, but it remains secondary to system correctness. The earlier FFT auto-ranging experiment caused visible freezing and frame-rate loss, so it was rolled back. Future visual work must preserve rendering performance.

20. Architecture at the End of Day 13

```
RTL-SDR V3
   |
   v
SDRWorker (QThread)
   |-- tune success/failure
   |-- connection/read error
   +-- IQ samples
          |
          v
Gain-compensated Hann FFT
          |
          +--> FFT and waterfall display
          |
          v
Local adaptive threshold
          |
          v
Resolution-aware peaks
          |
          v
Temporal confirmation
          |
          +--> signal table / history / FeatureStore
          |
          +--> occupancy and measurement buffer
                         |
                         v
                  median survey metrics
                         |
                         v
           deterministic SMART decision engine
                         |
                         v
       recommendation + runner-up + margin + confidence
                         |
                         v
              heatmap / auto-tune / report UI
```

21. Engineering Interpretation

Day 13 is a major project milestone. The system is no longer only feature-rich; it now has explicit controls for thread ownership, measurement quality, detector locality, temporal consistency, state freshness, deterministic ranking, and regression protection.

Before Day 13	After Day 13
SDR operations could block the GUI	SDR I/O and tuning execute in a dedicated worker thread
A tune transient could become a survey result	Validated multi-frame median measurements represent each point
Detection relied on one global baseline	Each frequency bin uses a locally adaptive threshold
Single-frame peaks were published	Peaks require temporal confirmation
SMART winner was shown without context	Runner-up, margin, and separation confidence are reported
Testing depended heavily on manual runs	65 automated tests protect the deterministic core
Popup was correct but visually raw	Structured professional report establishes the UI standard

22. Traceability - Selected Day 13 Commits

Commit	Milestone
e8d35fa	Move SDR acquisition and tuning off GUI thread
97a07fd	Add validated median RF measurement aggregation
55fdb5e	Use median post-tune measurements for automated surveys
57bcca9	Keep peak separation consistent across FFT resolutions
8c0c4a2	Add local adaptive noise floor estimator
56115e1	Integrate adaptive local threshold into live detection
2777ea6	Add gain-compensated Hann FFT processing
d06b7eb	Integrate temporal peak confirmation into live processing
566ba2a	Define deterministic SMART tie-breaking policy
07a4302	Add normalized decision confidence policy
9be9e9b	Redesign survey results as structured report

The repository records 54 incremental Day 13 commits. The table above lists the milestone commits rather than every stabilization and test-only checkpoint.

23. Outstanding Work

Item	Reason	Recommended Timing
Apply the approved report style to the main window	Current panels still use mixed spacing and visual hierarchy	Next UI phase, one panel at a time
Calibrate power reporting	RTL-SDR values are relative dB, not traceable dBm	Before claiming calibrated RF measurements
Validate detector across multiple RF bands	FM-band behavior alone is not a complete receiver validation	After UI baseline is stable
Add repeated-survey uncertainty metrics	Current confidence is score separation, not statistical certainty	After collecting repeat-run data
Review signal-type classifier	Frequency-band labels remain rule based	Future classification milestone
Add exportable structured survey data	CSV/JSON would improve reproducibility and report generation	After decision schema is frozen
Create operator README and setup guide	Hardware, driver, and macOS setup should be reproducible	Before final release

24. Recommended Next Sequence

Step	Controlled Deliverable	Exit Criterion
1	Freeze Day 13 logic baseline	Tests remain at 65/65 and live five-point survey passes
2	Extract shared UI palette and style constants	No functional changes and popup appearance is unchanged
3	Modernize the left survey-control panel	Spacing, labels, inputs, and buttons match the report style
4	Modernize right-side signal/status panels	Tables remain readable and live update rate is unaffected
5	Refine the central graph container	No auto-range loop, freezing, or measurable frame-rate loss
6	Run full regression and hardware acceptance test	Automated suite passes and repeated RTL-SDR survey is stable

25. Conclusion

Day 13 completed the reliability and measurement-quality foundation needed before broad UI work or additional cognitive features. The analyzer now keeps SDR operations off the GUI thread, aggregates stable post-tune measurements, detects peaks against a local RF baseline, confirms signals across time, prunes stale state, and produces deterministic SMART recommendations with transparent score separation.

The final automated result is 65 passing tests, supported by repeated live RTL-SDR V3 surveys and clean shutdown checks. The new Survey Analysis Report is both a completed UI milestone and the approved design reference for the rest of the application.

The correct next direction is controlled, application-wide UI standardization followed by multi-band validation and calibrated measurement work. Changes should continue in the same Day 13 pattern: one bounded milestone, automated regression, live hardware inspection, then commit and push.

Day 13 Acceptance Item	Final Status
GUI remains responsive during SDR acquisition and tuning	PASS
Survey measurement requires validated multi-frame aggregate	PASS
Adaptive detection integrated and regression tested	PASS
SMART ranking deterministic and explainable	PASS
65 automated tests pass	PASS
Repeated RTL-SDR V3 surveys complete	PASS
Professional results-report style approved	PASS